Response to Reviewers' Comments

We sincerely thank the Associate Editor and all reviewers for their helpful comments and suggestions. We have revised the manuscript and included here a point-by-point response. The main changes are the following:

1) ...
2) ...

Major changes are highlighted in blue in our revised manuscript. The references like equation numbers refer to those in the manuscript, except for those prefixed by "R".

Response to Reviewer 1

Recommendation: Major Revision

Comment 1.1. *A lot of comments.*

Response 1.1. Thank you for your comment. Please always use the formal "thank you" instead of "thanks". Please always start your response politely and in a conciliatory tone even if you are frustrated with the comment. We can be firm, friendly and respectful. After all, the reviewer has sacrificed his or her precious time to read our manuscript and provide comments, which he or she is not obligated to do.

Using the xr package, I can reference Section I, Fig. 2a and (1) of the main paper tips.tex. To do this, you need to build response.tex in the same folder containing tips.aux.

Next, I cite [13], [25], which are in the main paper. The references displayed are from tips.aux and tips.bbl thanks to xr. Finally, the catchfilebetweentags package allows me to quote the following from Section IX in tips.tex:

> This ensures that whatever changes you make in the tex file are reflected in the response. Examples given below are used in response.tex.

Note that the tag name cannot include underscores. Another quote from tips.tex:

> Other examples are as follows:
>
> $$\Lambda_{\mathrm{SGLR}}(k,t) = \max\left\{\Lambda(k,t), \Lambda_n(k,t)\right\}, \tag{3}$$
> $$\tau_{\mathrm{SGLR}}(b) = \min\{\tau(b), \tau_n(b)\}, \tag{4}$$
> $$\tau_{\mathrm{W-SGLR}}(b) = \min\{\widetilde{\tau}(b), \widetilde{\tau_n}(b)\}, \tag{5}$$
> $$\Phi_i = \tau_{\mathrm{W-SGLR}}(b) + i^2, \text{ for } i = 1, \ldots, N,$$
> $$\widetilde{\mathbf{C}} = \mathrm{one\_hot}\left(\widetilde{\mathbf{L}}\right), \tag{6}$$
> $$\mathbf{x} = \mathbf{B}^{\mathsf{H}}\boldsymbol{\Sigma}\mathbf{M}^{\mathsf{T}}\boldsymbol{\Phi}\mathbf{M}\mathbf{A}^{\mathsf{T}},$$
> $$d = D(p_X \,\|\, p_Z).$$

The equation numbers shown here are the same as those in the main paper, which is great!

Comment 1.2. *Do something.*

Response 1.2. Thank you for your suggestion. I will now cite some additional references that appear only in the bibliography for this response using [R-1]–[R-3]. These are not in the main paper. The steps to be taken are:

i) latex response
ii) bibtex lref.aux
iii) latex response
iv) bibtex lref.aux
v) copy lref.aux to lref-response.aux, and lref.bbl to lref-response.bbl
vi) latex response
vii) latex response
viii) latex response

These steps are automated in the lcite_.bat, lcite_.ps1, or lcite_.sh file. **If anyone can figure out a better solution, please let me know.** If you are using Overleaf, you need to copy the lref-response.aux and lref-response.bbl files to the Overleaf project folder.

Response to Reviewer 2

Recommendation: Major Revision

Comment 2.1. *Why does this hold?*

Response 2.1. Thank you for your question. Please refer to the following equation:

$$f(x) = \sum_{i>0} y_i. \tag{R1}$$

Note the prefix "R" in the equation that is in this response. From (R1) and (3) in the main paper, the result follows immediately.

Comment 2.2. *A lot of comments.*

Response 2.2. Thank you for your comment. Repeated quote from tips.tex:

> Other examples are as follows:
>
> $$\begin{aligned} \Lambda_{\mathrm{SGLR}}(k,t) &= \max\left\{\Lambda(k,t), \Lambda_n(k,t)\right\}, && (3)\\ \tau_{\mathrm{SGLR}}(b) &= \min\{\tau(b), \tau_n(b)\}, && (4)\\ \tau_{\mathrm{W-SGLR}}(b) &= \min\{\widetilde{\tau}(b), \widetilde{\tau_n}(b)\}, && (5)\\ \Phi_i &= \tau_{\mathrm{W-SGLR}}(b) + i^2, \text{ for } i = 1, \ldots, N, \\ \widetilde{\mathbf{C}} &= \text{one\_hot}\left(\widetilde{\mathbf{L}}\right), && (6)\\ \mathbf{x} &= \mathbf{B}^{\mathsf{H}}\boldsymbol{\Sigma}\mathbf{M}^{\mathsf{T}}\boldsymbol{\Phi}\mathbf{M}\mathbf{A}^{\mathsf{T}}, \\ d &= D(p_X \,\|\, p_Z). \end{aligned}$$

The equation numbers shown here are the same as those in the main paper, which is great! The command \revision takes care of this. But it has limitations. See the following examples.

We remark that it can be shown that

$$\Lambda(k) = \int_{\Omega} \tau(b) \,\mathrm{d}\nu(b). \tag{R2}$$

Comment 2.3. *Some other comments.*

Response 2.3. Thank you for your comment. Please refer to Fig. 2. We provide the following example, reproduced here:

> **Example 1.** *Consider the subset of vertices V_0 of an unweighted directed graph G. Suppose that we consider the adjacency matrix $\mathbf{A}_G$ as the graph shift operator for G. The adjacency matrix $\mathbf{A}_{H_0}$ of the subgraph H_0 shown is a submatrix consisting of the even-indexed rows and columns of $\mathbf{A}_G^2$. Any filter shift invariant with respect to (w.r.t.) $\mathbf{A}_{H_0}$ can be viewed as a polynomial of $\mathbf{A}_G^2$ composed with a projection to the vertices of H_0. See [2] for more details.*

Note the use of \xrcounterref to set the numbering of environments correctly. **If you know how to make this automatic, let me know.**

Comment 2.4. *Some other comments.*

Response 2.4. We provide the following theorem, reproduced here:

> The following is another theorem.

Theorem 0. *The limit posterior covariance of* $f(v_0, \mathbf{t})$ *over* $\mathcal{T}_0$ *converges:*

$$\lim_{M_0 \to \infty} \int_{\mathcal{T}_0} \mathrm{var}(f(v_0, \mathbf{t}) \,|\, \mathbf{y}(M_0)) \, \mathrm{d}\tau(\mathbf{t}) = \int_{\mathcal{T}_0} \mathrm{var}(f(v_0, \mathbf{t}) \,|\, \mathbf{z}) \, \mathrm{d}\tau(\mathbf{t}). \tag{9}$$

Note the use of \xrcounterref to set the numbering of environments correctly. **If you know how to make this automatic, let me know.**

Additional References

[R-1] D. Acemoglu, M. A. Dahleh, I. Lobel, and A. Ozdaglar, "Bayesian learning in social networks," *Review of Economic Studies*, vol. 78, no. 4, pp. 1201–1236, Mar. 2011.

[R-2] P.-N. Chen and A. Papamarcou, "Error bounds for parallel distributed detection under the Neyman-Pearson criterion," in *Proc. IEEE Int. Symp. Inform. Theory*, Mar. 1995, pp. 528–533.

[R-3] T. M. Cover, "Hypothesis testing with finite statistics," *Annals of Mathematical Statistics*, vol. 40, no. 3, pp. 828–835, 1969.